



Constraining turbulence in the ICM with the SZ effect

by

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CURRENT DEGREES



Candidate's ORCID

*A thesis submitted to the Victoria University of Wellington in partial
fulfillment of the requirements for the degree of Doctor of Philosophy*

Te Herenga Waka - Victoria University of Wellington

2027

Abstract

Gas motions. SZ effect Pressure fluctuations. Stable methodology. modeling. simulation and some interferometry.

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Acknowledgments

Praise to Allah. Appreciation to my parents. Thanks to my supervisors.

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- Dedications (if applicable)
- Table of Contents
- List of Figures and Tables
- List of Abbreviations used in the thesis
- Main text of the thesis
- Bibliography or List of References
- Appendices

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For F.N.E. May she live forever.

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List of Abbreviations and Symbols

Abbreviations

AC	Alternating Current
AFM	Atomic Force Microscopy/Microscope
<i>etc.</i>	<i>etc.</i>

Symbols

$\hat{\rho}$	Density operator
<i>etc.</i>	<i>etc.</i>

Chapter 1

Scientific Landscape

“If you wish to create a Universe from scratch, the final ingredient would be clusters of galaxies.”

— [Afiq Hamid 2025]

1.1 Galaxy Clusters

It is said that mass and time are among the most important parameters in all of astrophysical science. This thesis is an exploration of the former, through the study of the most massive virialized objects known to exist in the Universe: galaxy cluster.

1.2 Observables

Galaxy clusters appear differently based on the way they are observed.

The following publication has been incorporated as Chapter 2.

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Chapter 2

Full title

Introduce the broad layout of the chapter.

2.1 Introduction

Add your text here.

Chapter 3

Conclusion

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Appendix A

Appendix

Write your appendix here. Following two are examples.

A.1 Name of Appendix-1

A.2 Name of Appendix-2

Appendix B

Example of Citations

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Appendix C

Example of Code

C.1 Find the greatest number from a list of numbers in *Python*

```
a=[1,2,3,4,6,7,99,88,999]
max= 0
for i in a:
    if i > max:
        max=i
print(max)
```


Appendix D

Example of Equations

$$E = mc^2 \tag{D.1}$$

$$a = b + c \tag{D.2}$$

$$x = y + z \tag{D.3}$$

$$\begin{aligned} a &= b + c \\ &\quad + d + e \end{aligned}$$

$$\sin x + \cos x = 1 \tag{D.4}$$

Appendix E

Example of Figures



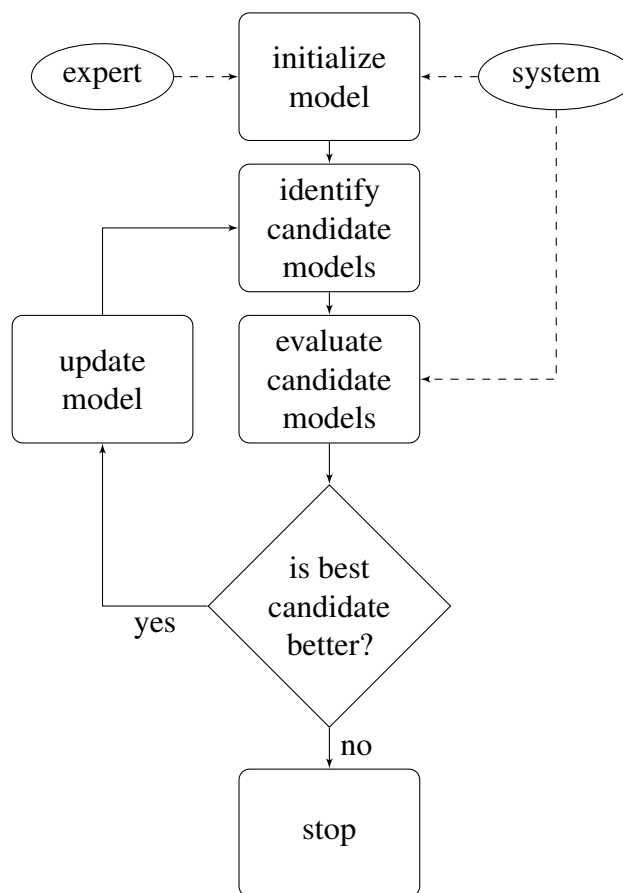
Figure E.1: The University Of Queensland

The Figure E.1 represents beauty of the UQ campus.

Appendix F

Example of Flow Charts

Figure F.1: Flow Chart



Flow chart F.1 is a simple example.

Appendix G

Example of Tables

Here is a really simple table G.1.

Table G.1: Name of the Australian Cities

Number	Name
1	Brisbane
2	Sydney
3	Melbourne
4	Canberra
5	Perth
6	Adelaide
7	Hobart
8	Darwin

Endquote goes here.

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